SPICE vs Spingarn on three crossbar models

Setup. Each case is an $N \times N$ crossbar with source voltages $E \in [-1, 1.5]$ V, per-cell $R_{\min}$ log-uniform in 1–10 kΩ and $R_{\max}$ in 50–200 kΩ, 5 Ω wires, 50 Ω sources and 1 kΩ loads. The cell is a tanh memristor alone (*memristor*), or a memristor in series with an ideal or a Shockley diode ($I_s = 10^{-9}$ A, $n = 1$) on a branch of its own through an internal node (*1D1R*). Every size has ten random circuits (seeds 0–9), solved by both solvers. SPICE is ngspice-47 with KLU and RELTOL 10^{-3}; it writes the ideal diode as uramp(V)/1 mΩ and the Shockley diode as a behavioural exp source. Spingarn keeps its port resistances fixed (no adaptive re-matching) and factorizes with CHOLMOD in its own (AMD) ordering, with $\alpha = 1$ on the memristor and 0.8 on 1D1R; zero-bias $\gamma = R_{\min}$ per memristor and, on 1D1R, the diode port $\sqrt{R_{\min} R_{\max}}$ per cell. Both stop on the same KCL test. Each run executed in its own process on one machine (Intel i5-11600KF, 31.8 GB, single-threaded BLAS); runtime is the whole solve. A run still solving after 1600 s was stopped and counts as failed to converge within the time limit. SPICE was not run at 1024×1024, where ngspice's KLU fails on these circuits regardless of time.

Results (medians over the ten circuits; Tables 1–2, Figures 1–5).

- *Memristor.* SPICE converged by plain Newton in 5 iterations at every size; Spingarn's median stayed at 19–22 iterations at every size up to 1024×1024. Spingarn was faster at every size: 3.9× at 16×16, 12× at 512×512 (3.69 against 44.2 s; 0.72 against 3.09 GB), and solved 1024×1024 in 15.9 s (range 15.5–16.3 s, 2.83 GB).

- *1D1R, ideal diode.* SPICE took a median of 8–16 Newton iterations, plain Newton in 59 of its 60 runs (one 16×16 circuit needed gmin stepping). Spingarn's iterations stay at 113–130 up to 128×128 and then grow about threefold per doubling of N (170, 496, 1596 at 256, 512, 1024). It ties SPICE at 16×16 and is 3.1× faster at 256 and 512 (90.6 against 279 s; 0.98 against 4.59 GB); at 1024×1024 it took 1169 s (1144–1233 s), every run within the cap.

- *1D1R, Shockley diode.* SPICE needed gmin stepping on every run (medians of 78–94 iterations) and was 1.4× slower than Spingarn at 16×16, 35× slower at 256×256 (284 against 8.2 s; 4.6 against 0.30 GB). On one 128×128 circuit ngspice reported success after 1255 iterations of gmin stepping, but its operating point failed the KCL test (relative residual 2.6×10^{-3}, above RELTOL), so it counts as not converged. At 512×512 the first circuit was still unsolved at the 1600 s cap; KLU's reordering had taken the process to a peak of at least 19 GB of RAM and 27 GB committed, leaving the 32 GB machine almost without free memory, so the other nine circuits were not run. Spingarn took the same iterations as with the ideal diode (158, 486, 1610 at 256, 512, 1024): 110 s at 512×512, at least 14.6× faster than SPICE, and 1445 s (1418–1479 s) at 1024×1024, every run within the cap.

- *Agreement.* The two solvers' word- and bit-line potentials differ by at most 1.2×10^{-10} V on the memristor and 2.2×10^{-7} V on the ideal diode (SPICE's 1 mΩ diode). On the Shockley diode they agree to 7.5×10^{-8} V or better except on two circuits where SPICE stopped with a looser KCL residual (within RELTOL), 6.6×10^{-7} and 7.8×10^{-6} V.

	Memristor			1D1R, ideal diode			1D1R, Shockley diode		
N	SPICE	Spingarn	ratio	SPICE	Spingarn	ratio	SPICE	Spingarn	ratio
16	0.0239	0.00619	3.9	0.0260	0.0278	0.9	0.0479	0.0344	1.4
32	0.0620	0.0104	6.0	0.0780	0.0623	1.3	0.180	0.0719	2.5
64	0.212	0.0405	5.2	0.328	0.256	1.3	1.47	0.329	4.5
128	0.984	0.156	6.3	1.87	1.12	1.7	$21.6^{9/10}$	1.16	18.6
256	5.39	0.775	7.0	23.5	7.52	3.1	284	8.16	34.8
512	44.2	3.69	12.0	279	90.6	3.1	>1600	110	>14.6
1024	–	15.9	–	–	1169	–	–	1445	–

Table 1: Median runtime (s) over ten circuits, and SPICE's median over Spingarn's (ratio > 1: Spingarn faster). A superscript k/m: only k of the m runs converged (a run stopped at the 1600 s cap counts as slower than all converged runs in the median). >1600: the run hit the cap. –: not run.

	Memristor		1D1R, ideal diode		1D1R, Shockley diode	
N	SPICE	Spingarn	SPICE	Spingarn	SPICE	Spingarn
16	5	22	8	113	94	110
32	5	20	8	124	82	114
64	5	19	9	130	78	115
128	5	20	10	129	$88^{9/10}$	115
256	5	20	12	170	88	158
512	5	20	16	496	t/o	486
1024	–	20	–	1596	–	1610

Table 2: Median iterations over ten circuits: Spingarn's, and SPICE's Newton iterations including fallback stages (gmin stepping). t/o: the run hit the 1600 s cap.

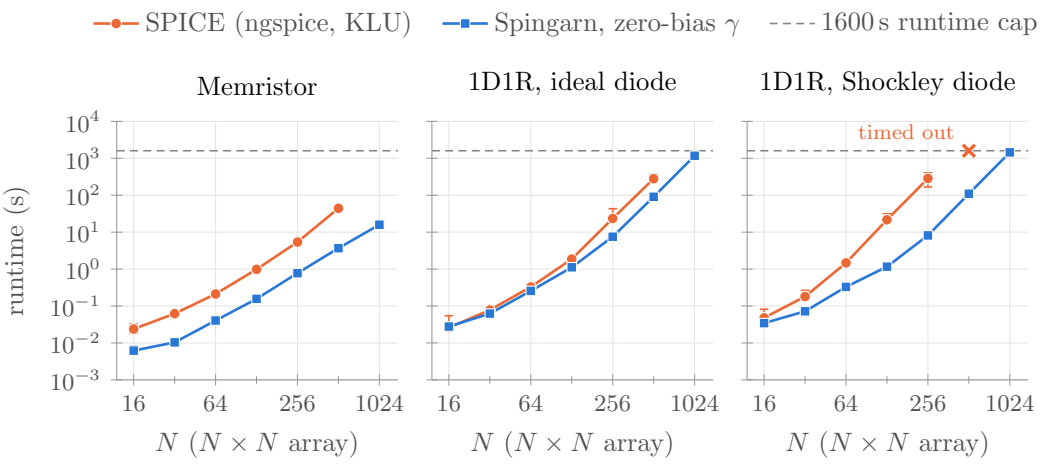


Figure 1: Runtime against array size, median over ten random circuits, bars from the fastest to the slowest converged run. Dashed: the 1600 s cap; a cross marks a size where SPICE hit it.

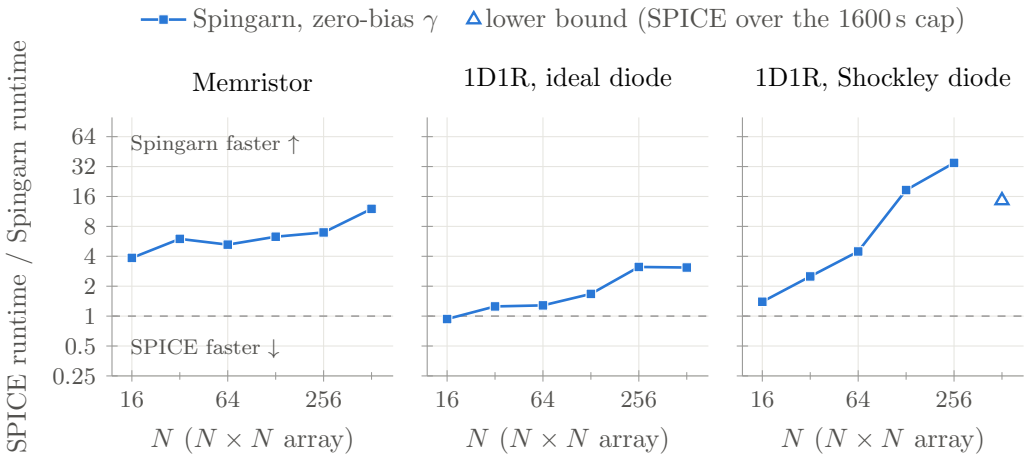


Figure 2: SPICE's median runtime divided by Spingarn's. Where SPICE hit the cap, the open triangle is the lower bound 1600 s over Spingarn's median.

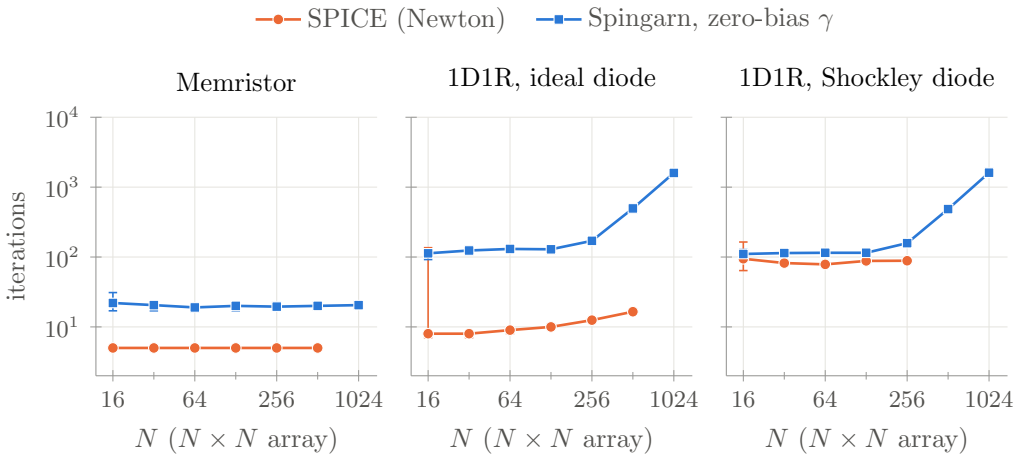


Figure 3: Iterations: Spingarn's, and SPICE's Newton iterations including fallback stages (gmin stepping), median and range over ten circuits.

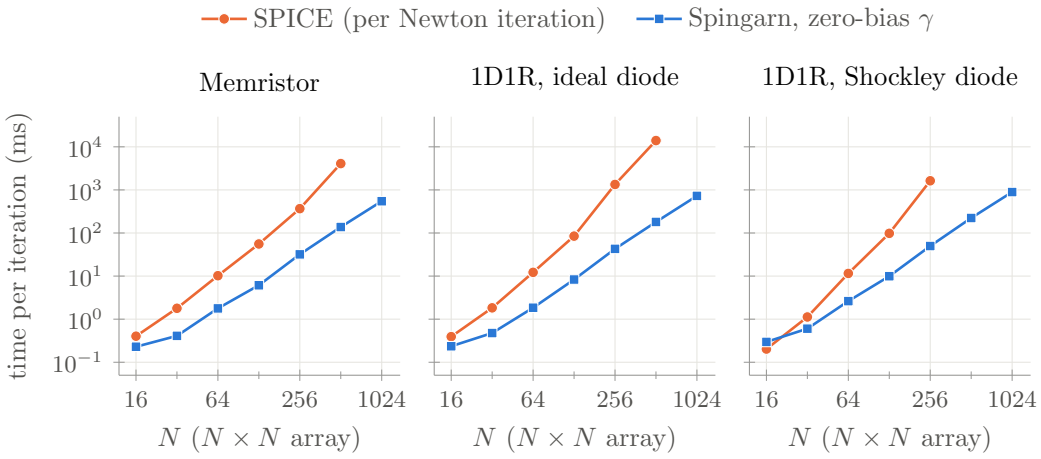


Figure 4: Time per iteration, one-time costs (netlist load and KLU reordering; Spingarn's assembly and factorization) excluded; median over converged runs.

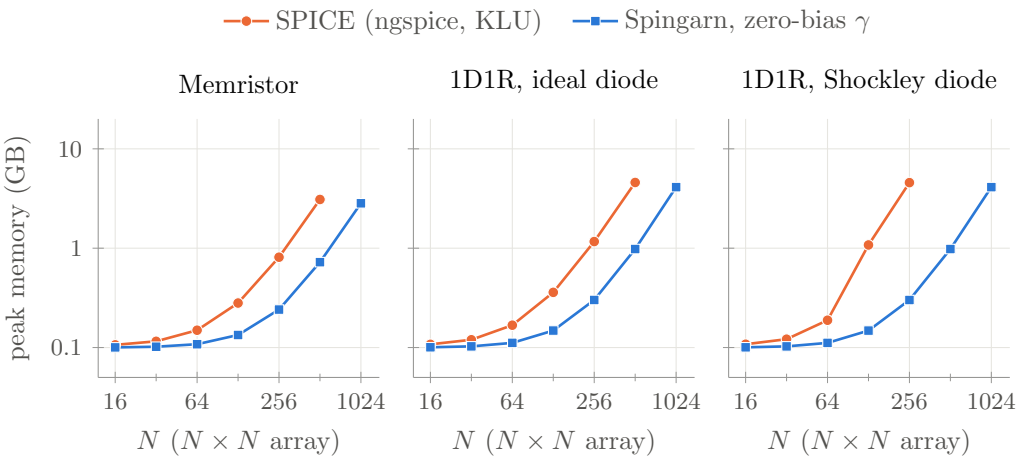


Figure 5: Peak memory of the process each run executed in, median over converged runs.